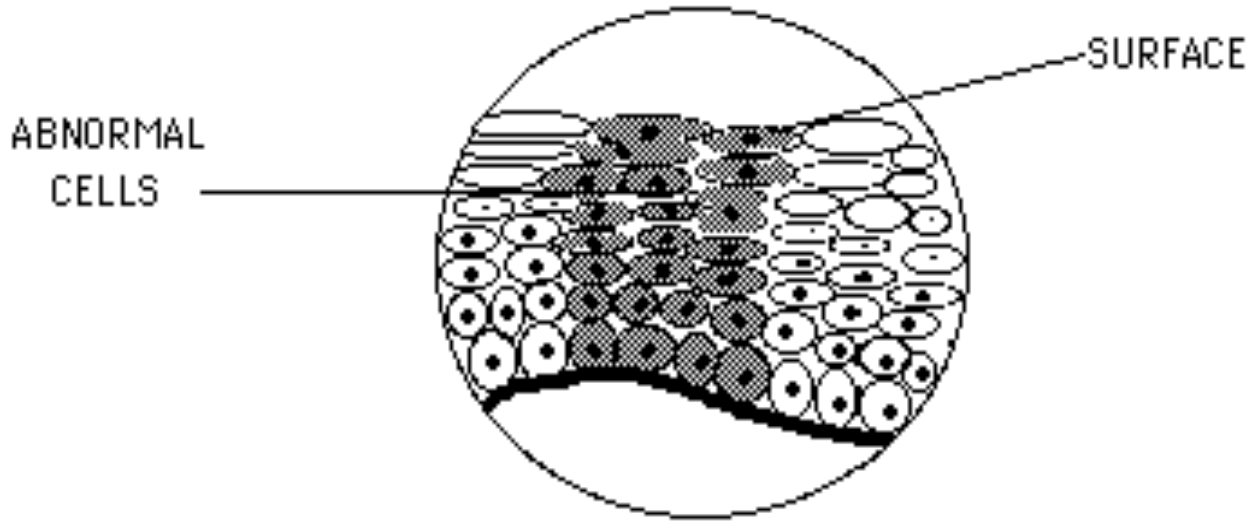
Premalignant lesion

Premalignant lesion

A clinically benign, histologically altered tissue that has a greater risk of malignant transformation than normal.

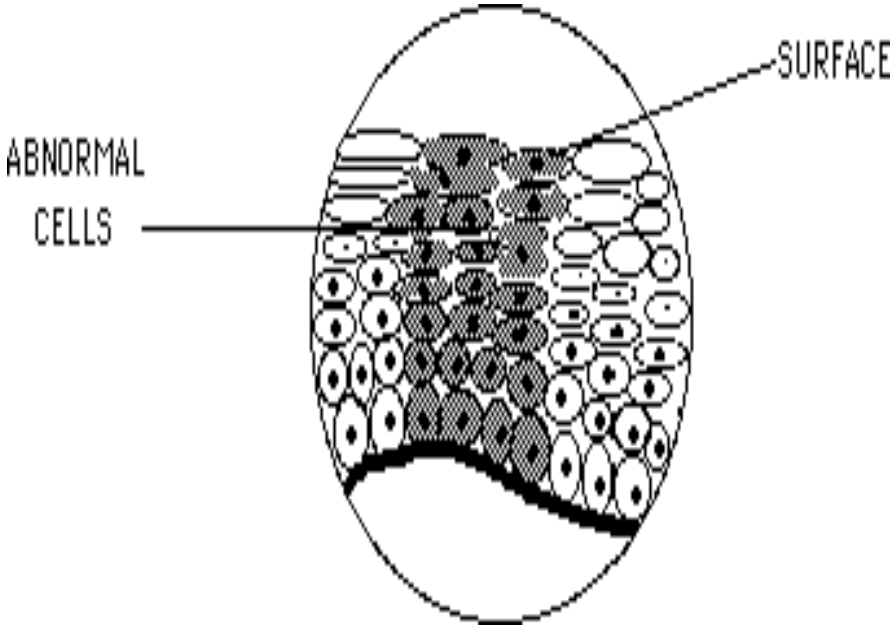
Premalignant lesion

1. Epithelial dysplasia
2. Intact basement membrane
3. No invasion to underlying C.T.
4. No metastasis



SEVERE DYSPLASIA, OR CARCINOMA-IN-SITU

Carcinoma insitu



Epithelial dysplasia

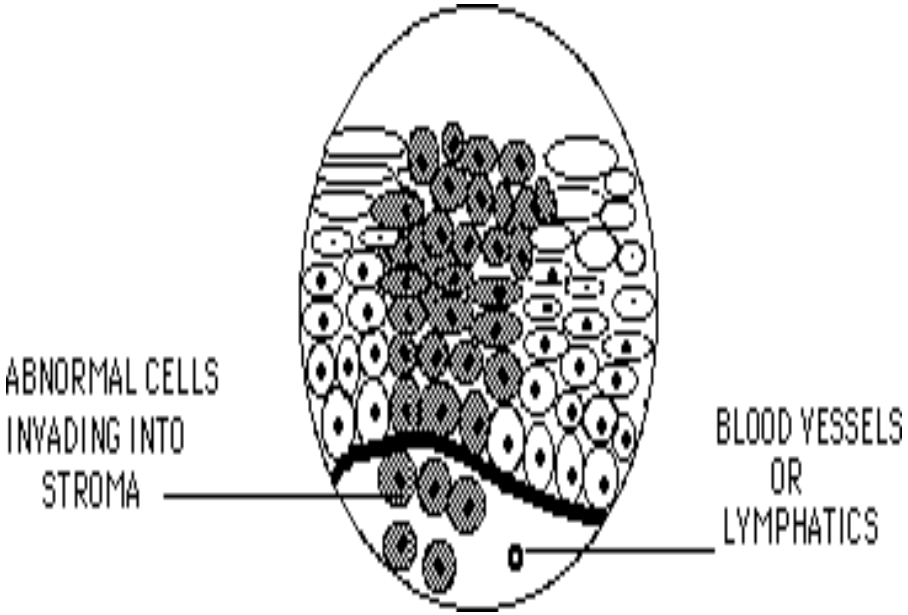
+

Intact basement membrane

=

Premalignant lesion

Invasive cancer



Epithelial dysplasia

+

Rupture of basement membrane

=

Invasive carcinoma

Premalignant lesion

➢ Types

*Leukoplakia

*Erythroplakia

*Actinic chelitis

* oral submucous fibrosis

* PVL (proliferative verrucous leukoplakia)

Epithelial dysplasia

- ***"Epithelial dysplasia"*** **= abnormal and disordered formation of epithelial tissue**
- ***"Atypia"*** **= abnormal nuclear features**

Criteria of epithelial dysplasia

A. Cytological (tissue/ cellular) changes

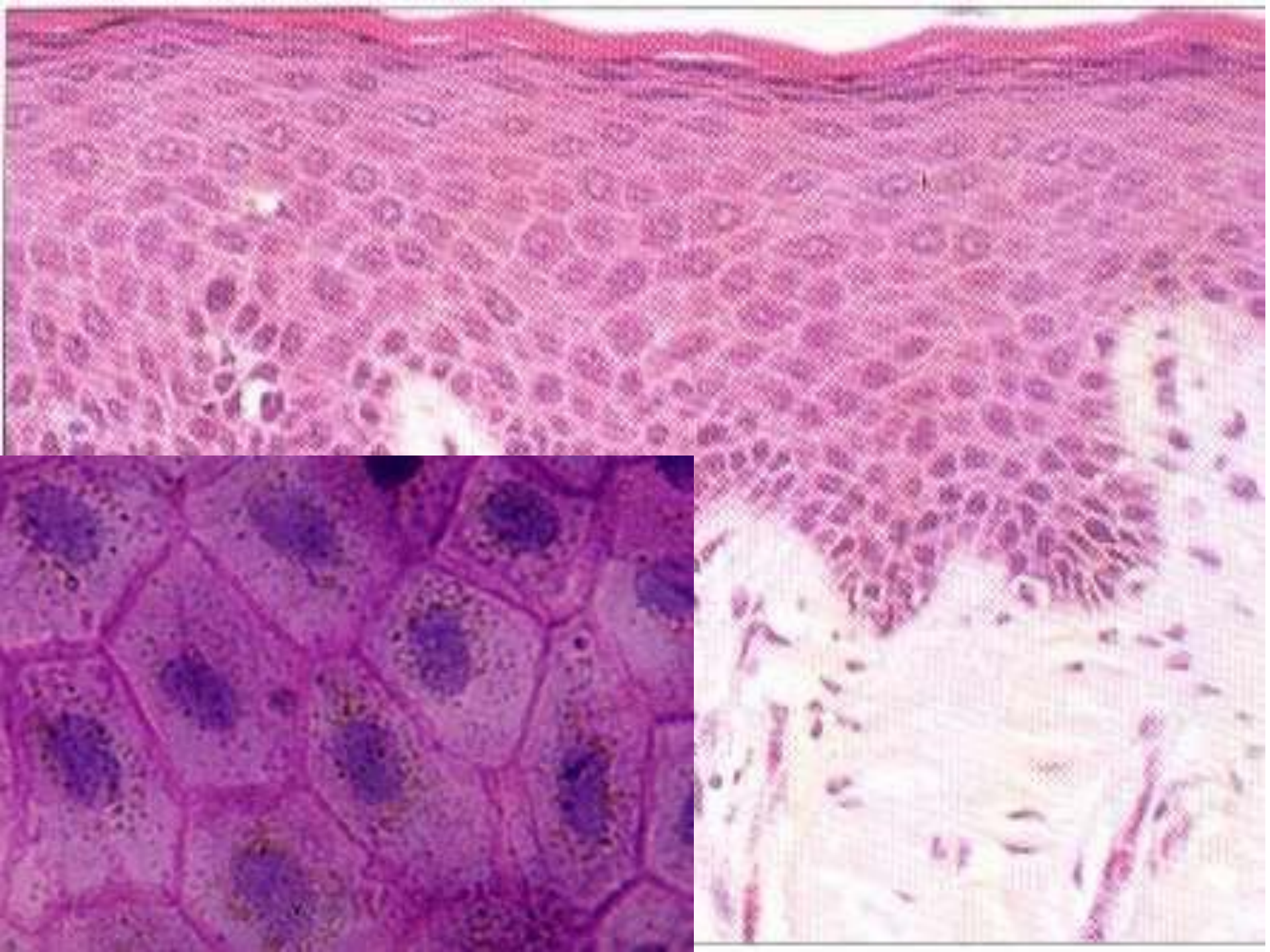
1) Basal cell hyperplasia.
2) Drop shaped rete pegs.
3) Disturbed polarity of basal cells or loss of cellular orientation.
4) Loss or reduction of intercellular adhesion (or cohesion).
5) Irregular epithelial stratification or disturbed maturation.

B. Nuclear changes

1) Nuclear hyperchromatism
2) Alteration (↑) in N/C ratio.
3) Increased and abnormal mitoses.
4) Dyskaryosis
5) Piokilokarynosis
6) Nuclear and cellular pleomorphism.
7) Prominent and enlarged nucleoli

C. Dyskeratosis.

1) individual cell keratinization
2) Keratin or epithelial pearls

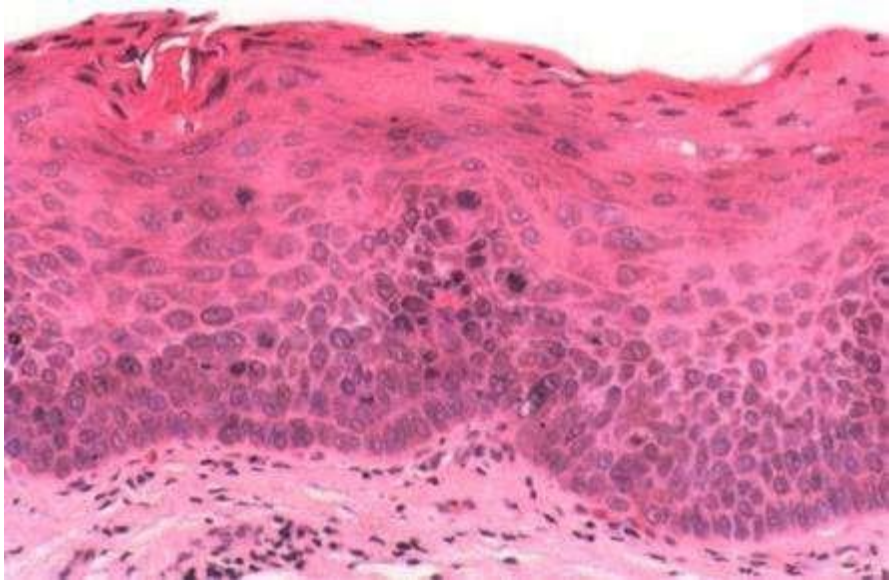


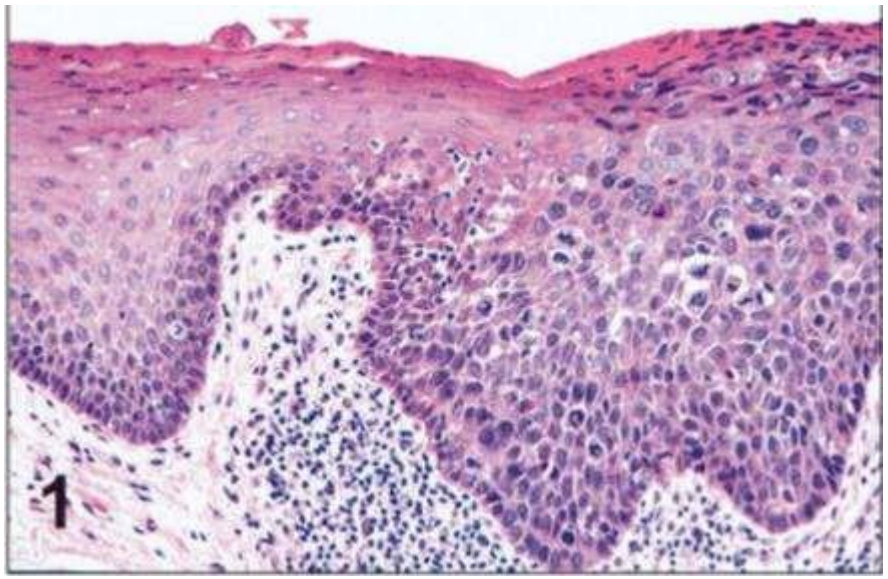
Criteria of epithelial dysplasia

A- Cytological (cellular) changes (evident at low–power magnification)

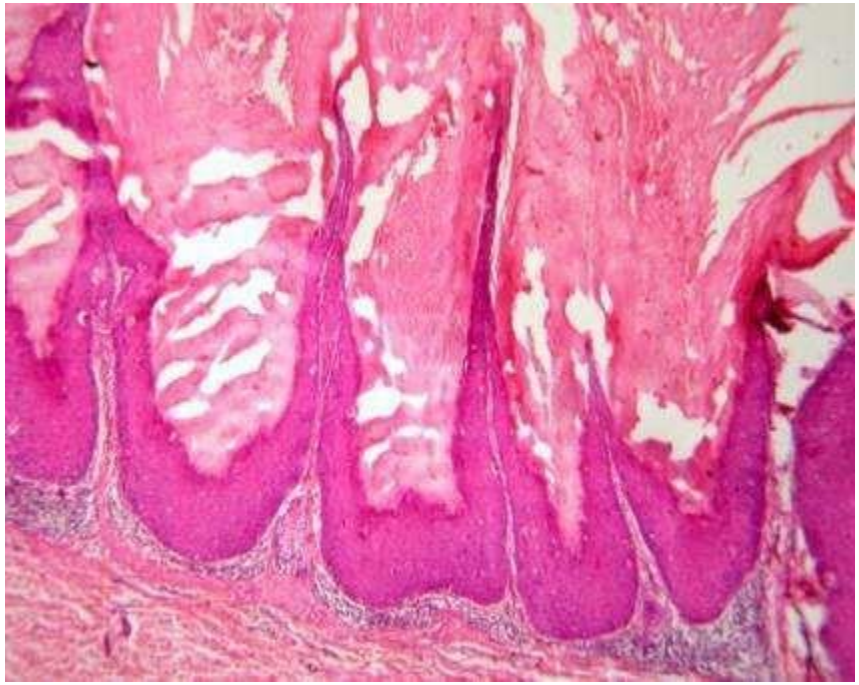
1.Basal cell hyperplasia.

The presence of several layers of basal cells & appearance of crowded cells. It is often associated with drop–shaped rete pegs.





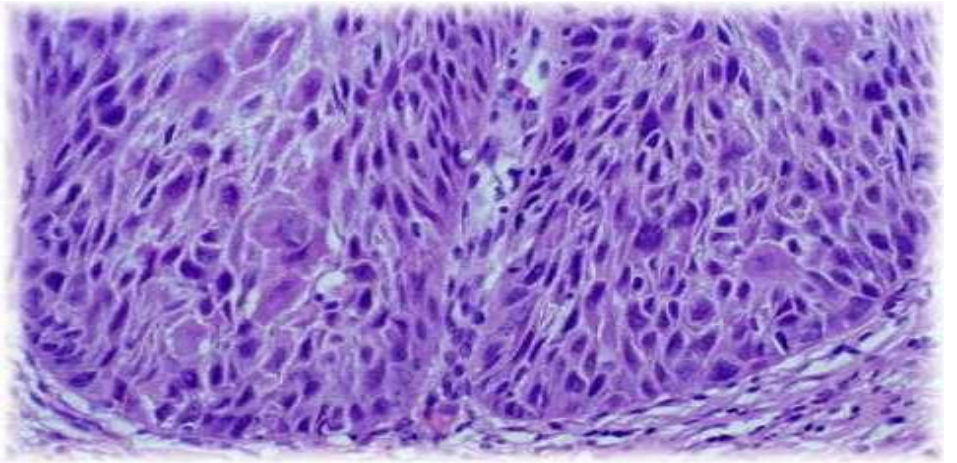
- **Drop shaped rete pegs→ The rete pegs are wide at their deepest part than they are more superficially.**

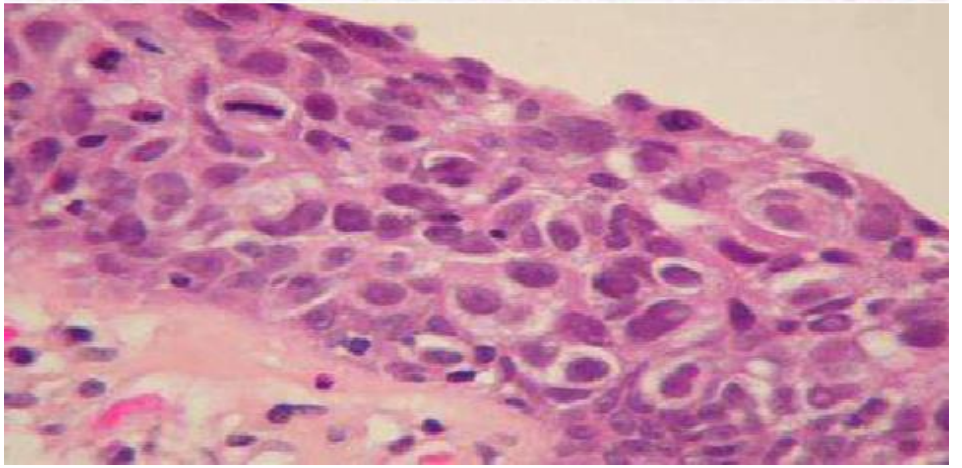


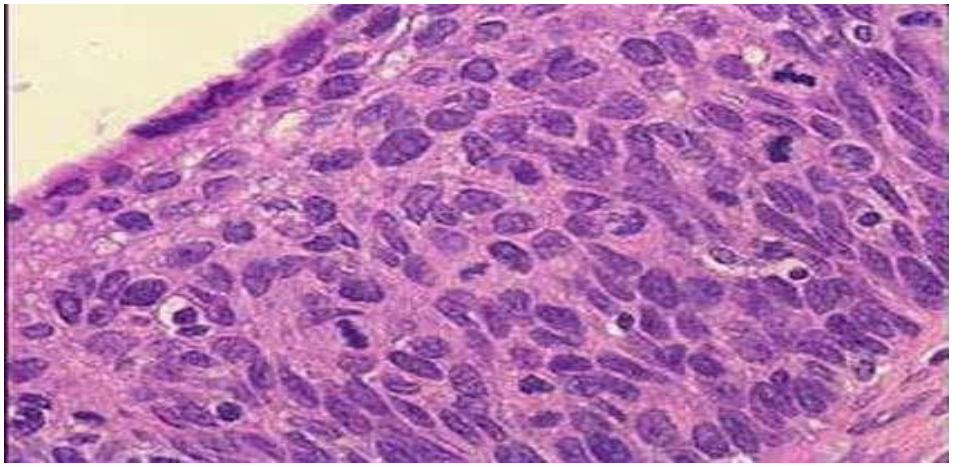


3.Disturbed (or loss) polarity of basal cells

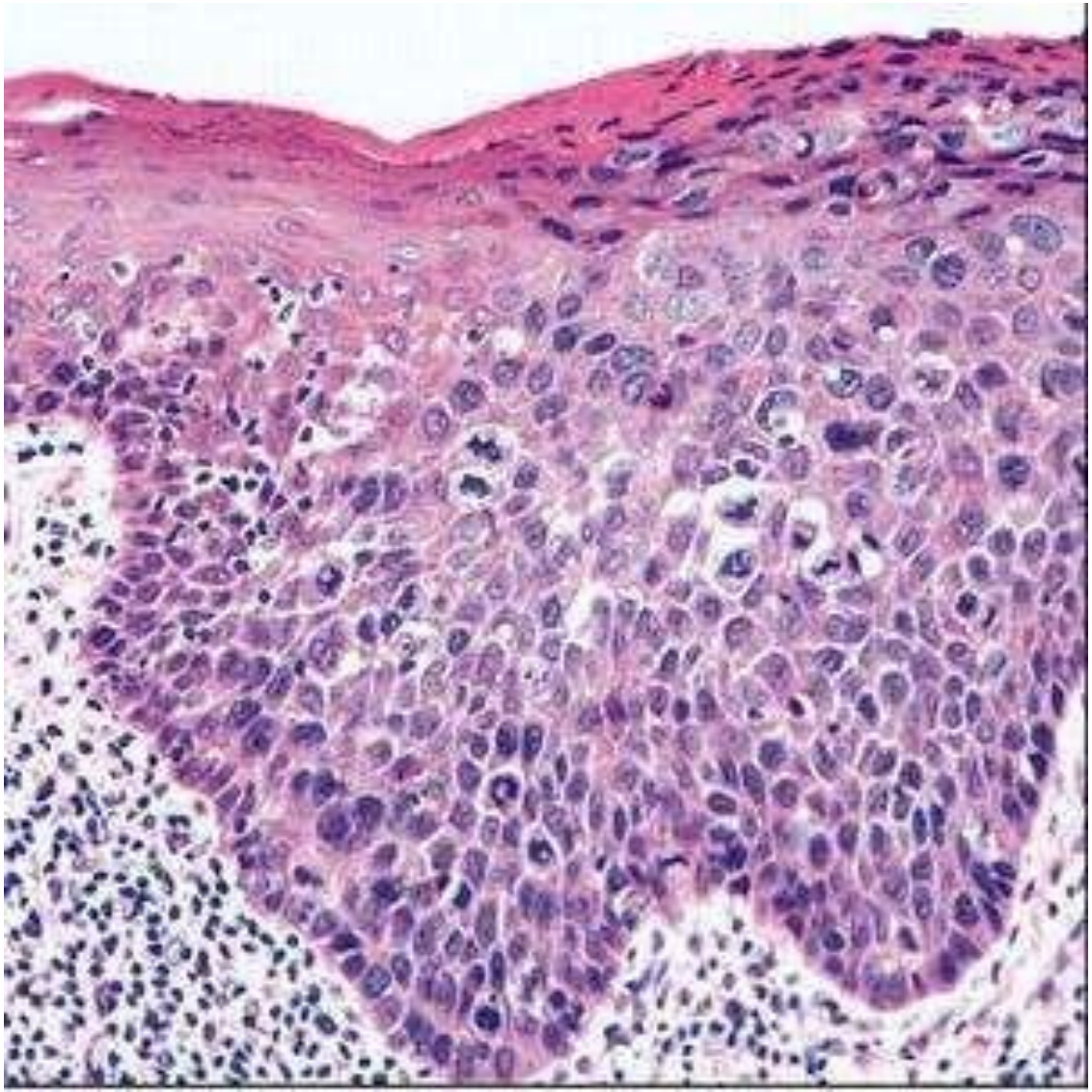
The basal cells have no definite long axis directed to the surface so nuclei have no polarity due to loss of hemidesmosome.



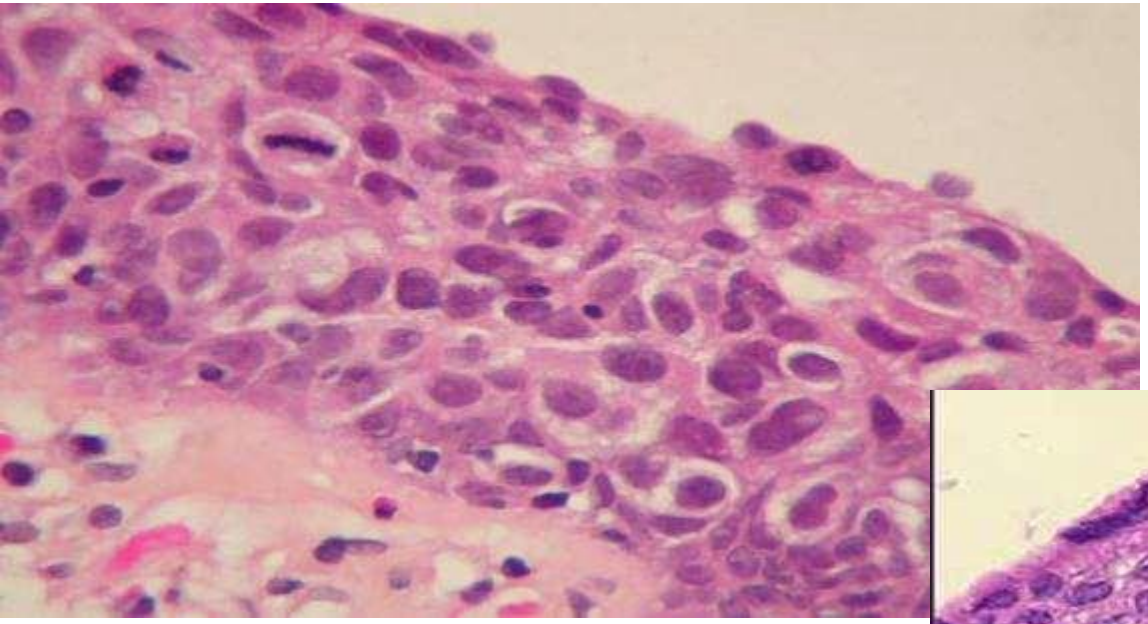


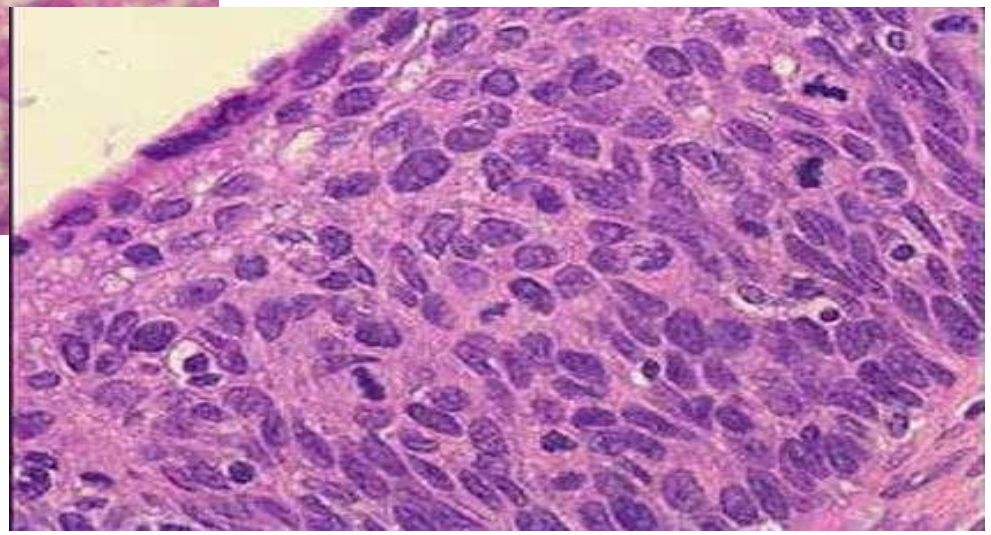


4. Loss or reduction of intercellular desmosomal junction → loss of cellular adhesion (or cohesion).

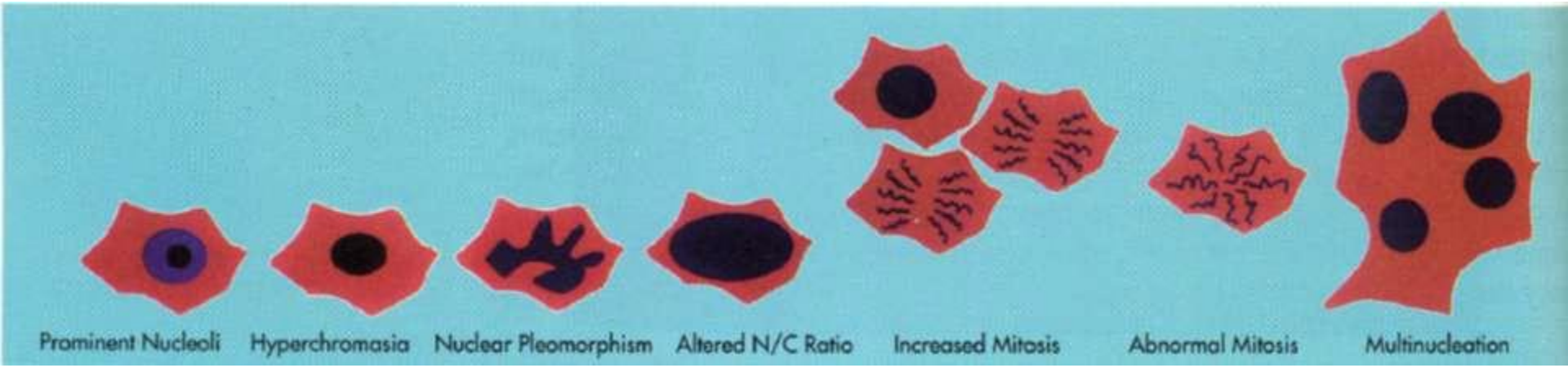


5. **Irregular epithelial stratification or disturbed maturation.** **The cells no longer show a proper sequence of morphological and maturational changes as they pass from the basal layer to the surface.**

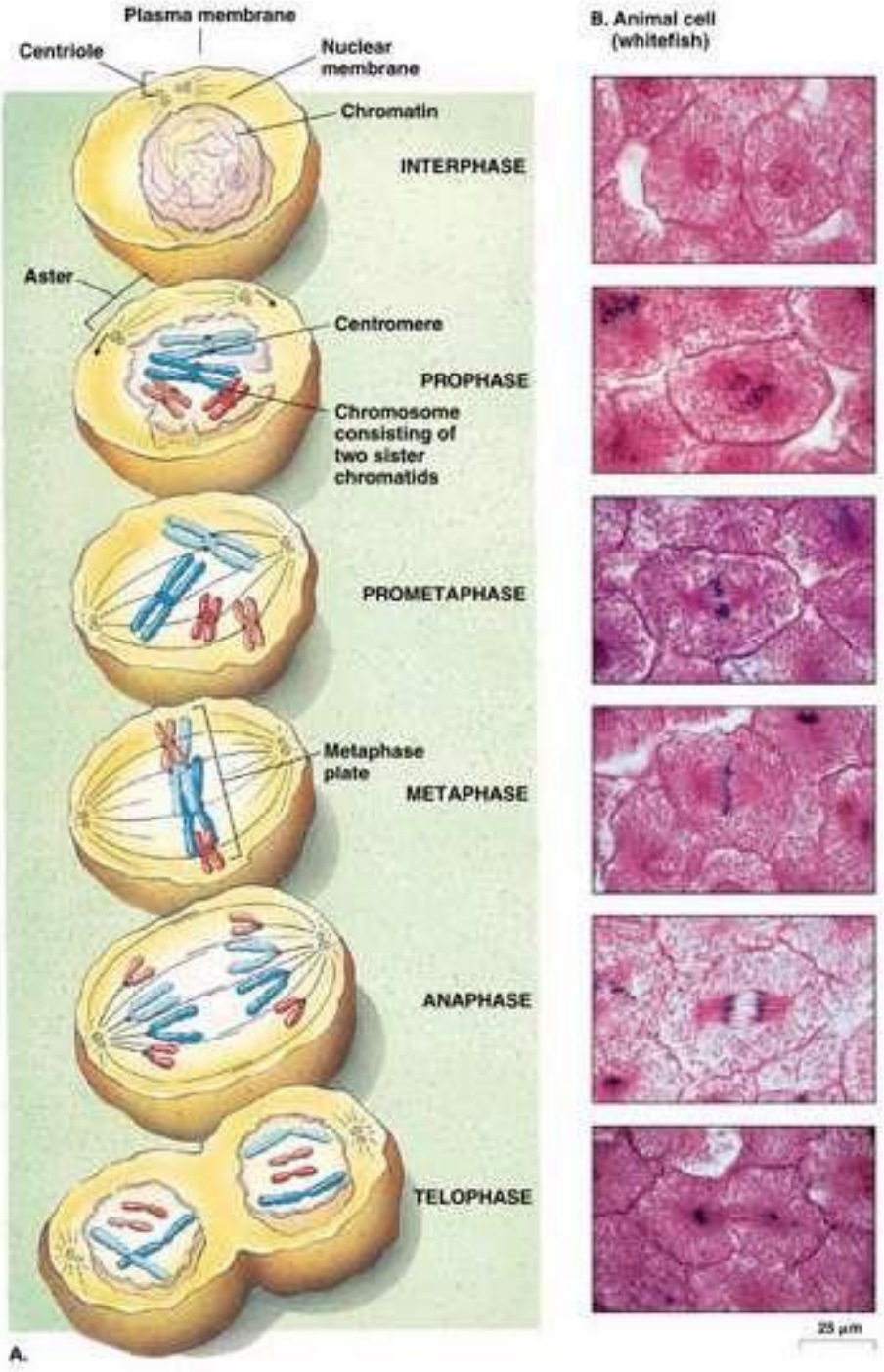




B- Nuclear changes (evident at high–power magnification)

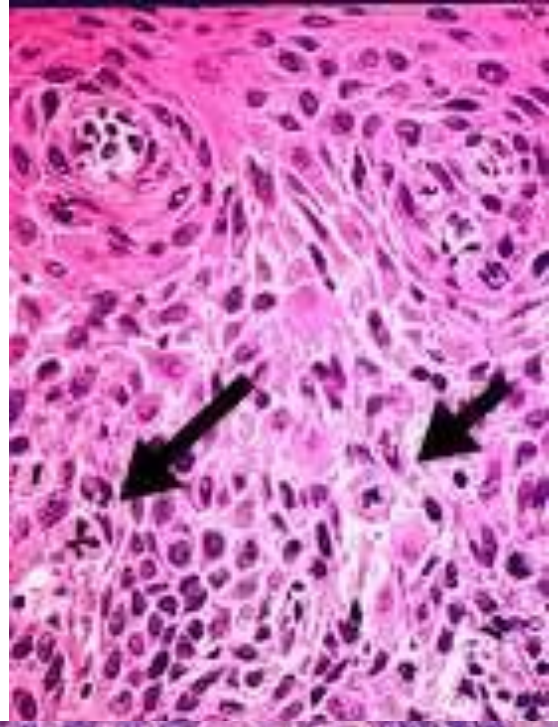


Normal mitosis

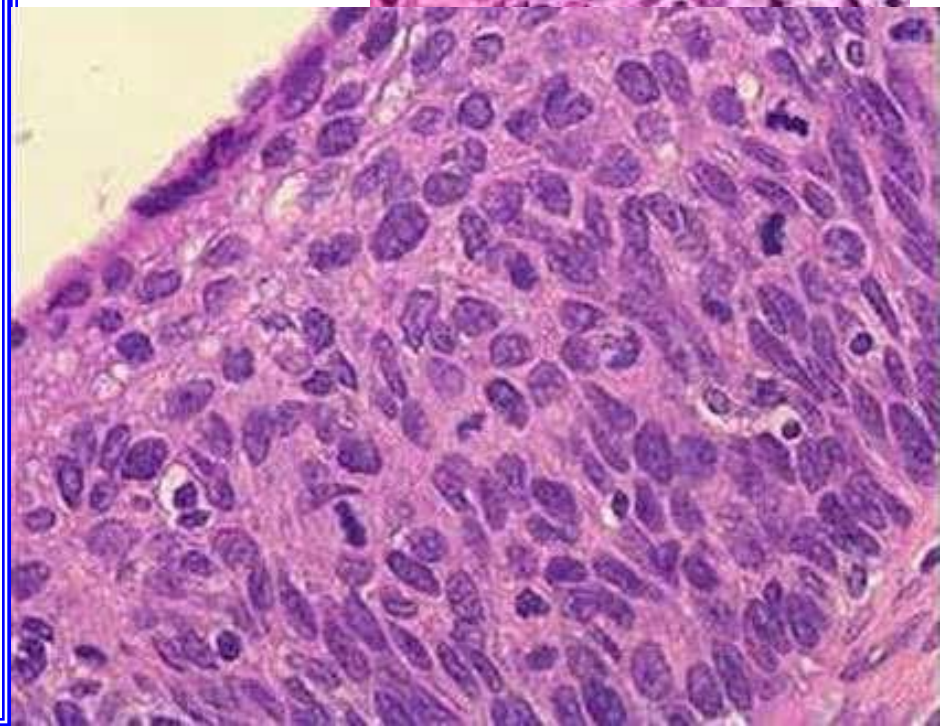


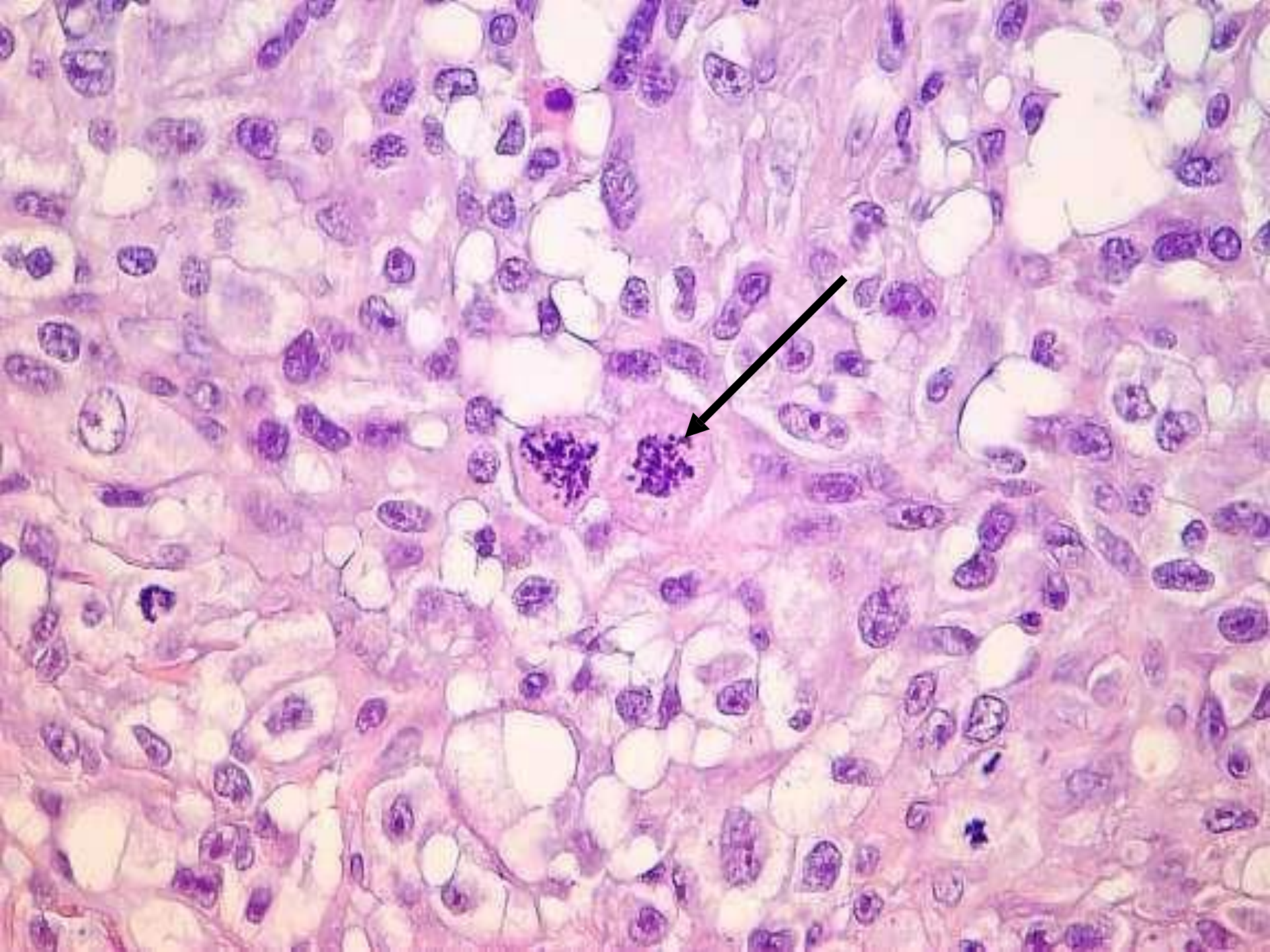
B- Nuclear changes (evident at high–power magnification)

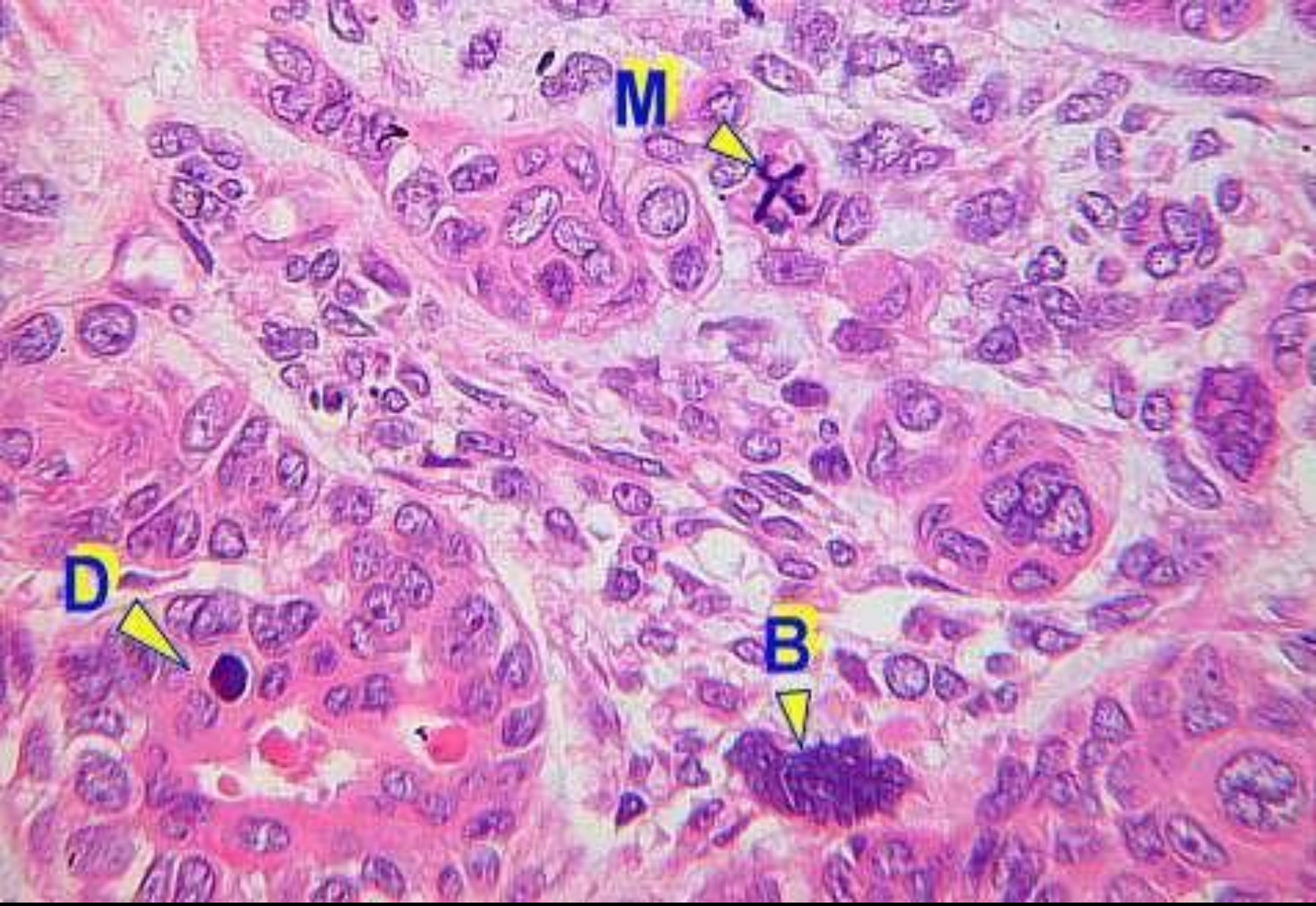
1) Increased mitoses and abnormal mitotic figure.



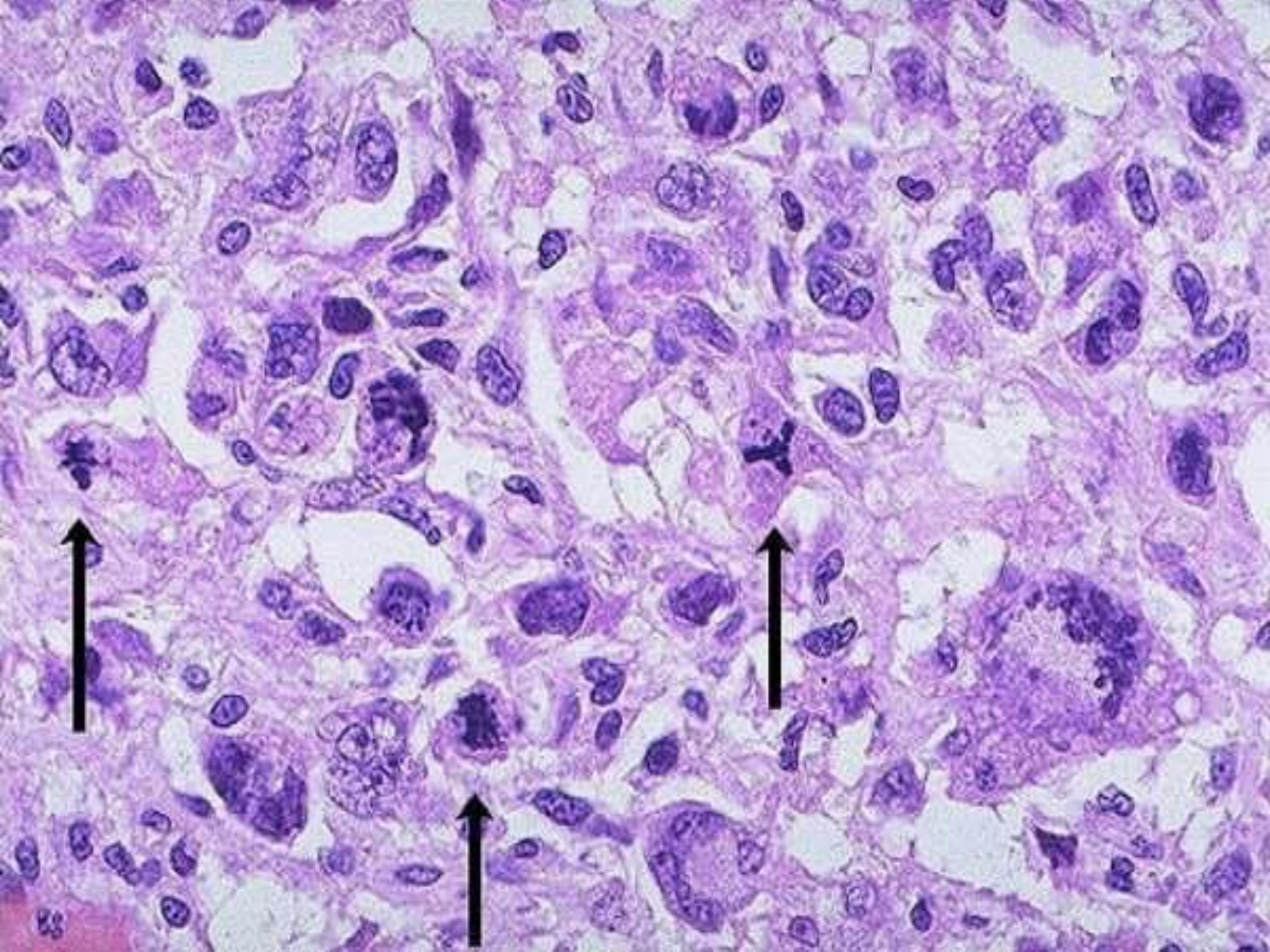
A. Increased in number
B. Abnormal location within the epithelium and may appear at all levels rather than in its usual basal location
C. Abnormal form/figure (tripolar or star shape or bizarre shape mitoses).





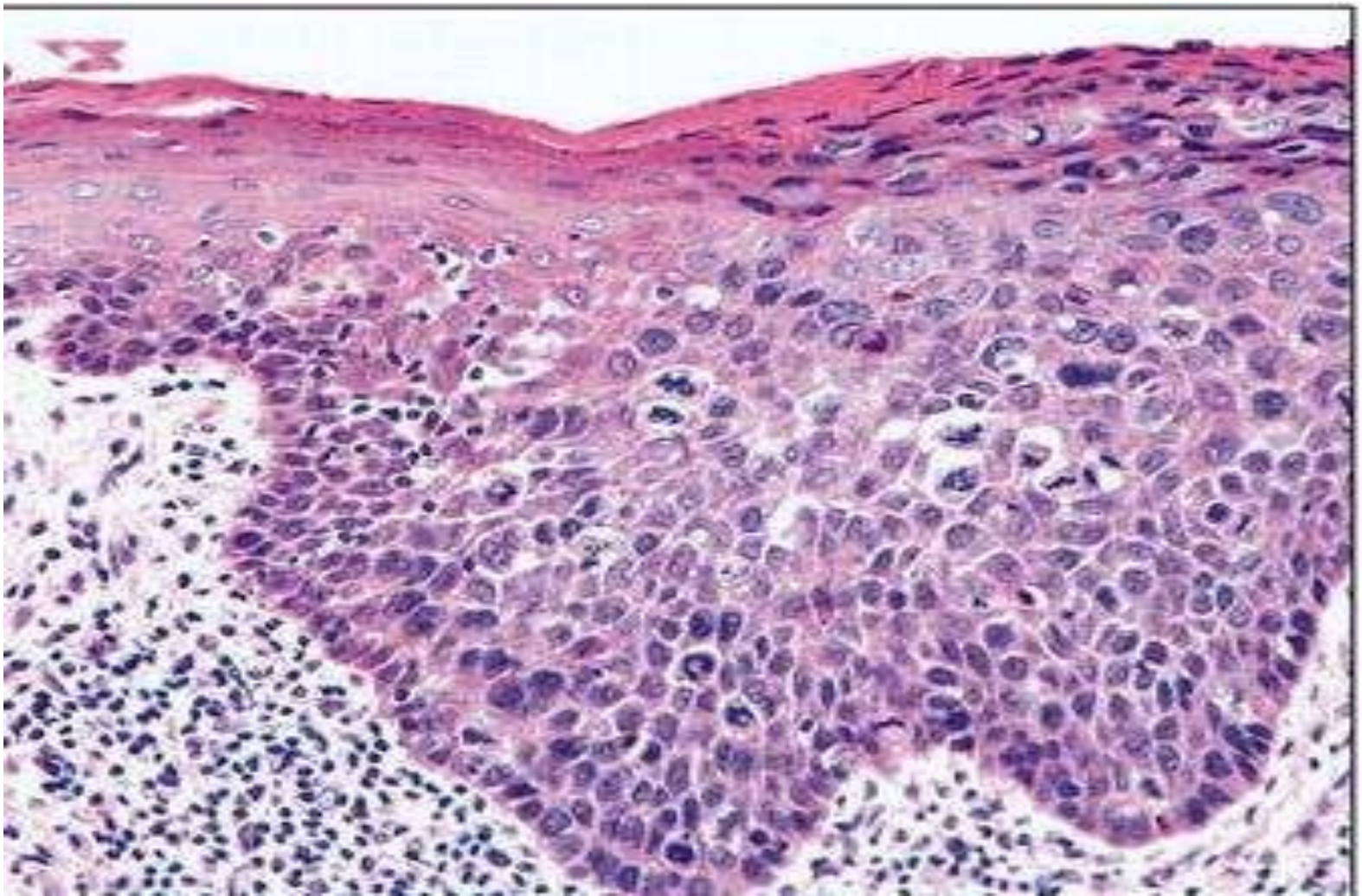


atypical mitoses (M), bizarre cells (B)



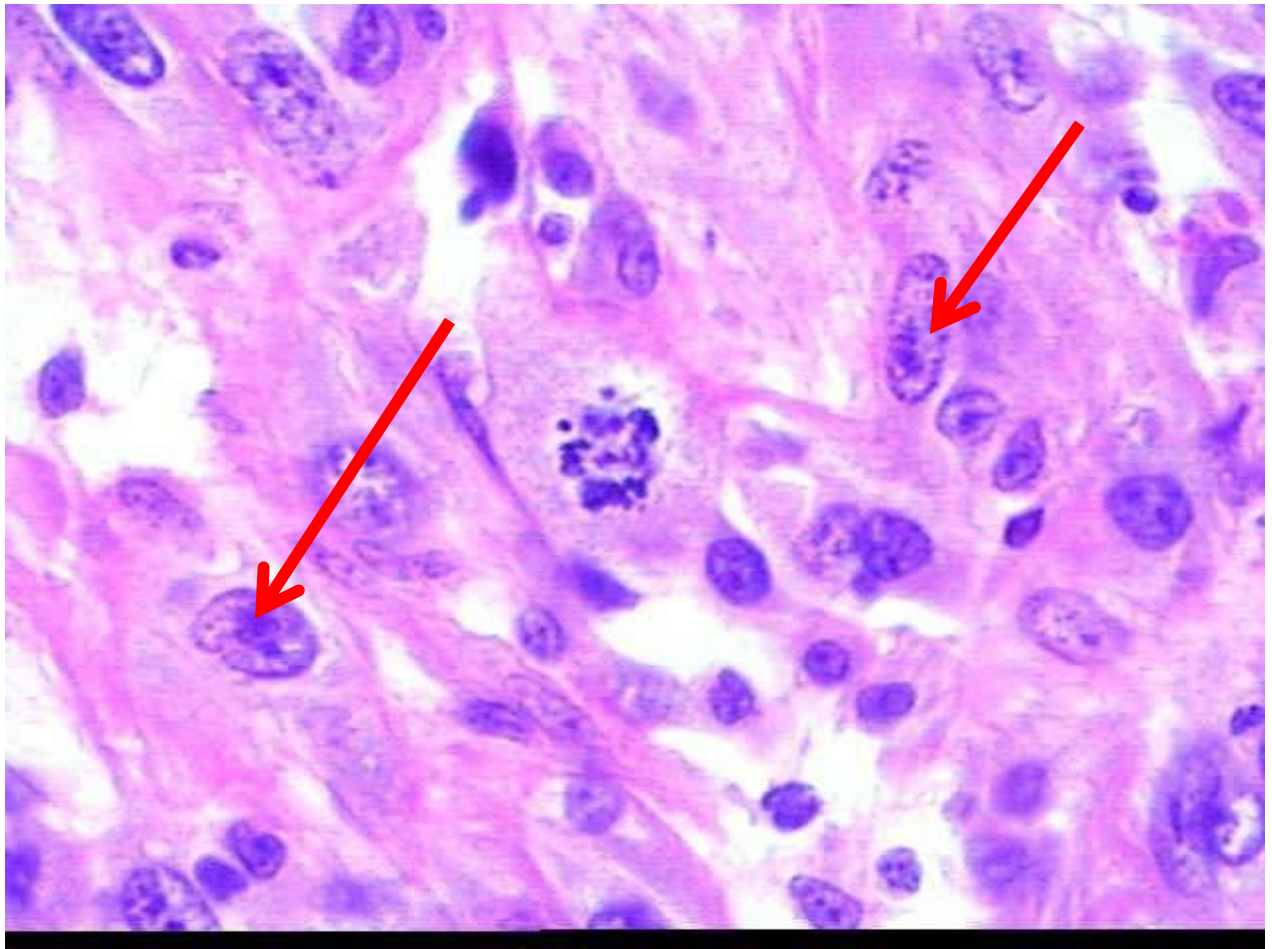
2) **Alteration (increase) in nuclear/cytoplasmic ratio** from 1:4 to 1:1 (N=nucleus ≈ 1/6--1/4 of cell).

3) **Nuclear hyperchromatism** (abnormally excessively dark–staining nuclei)

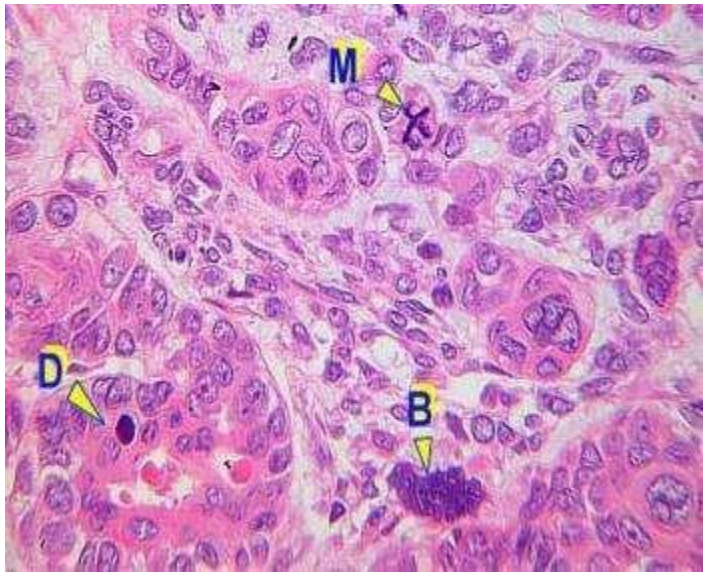


4. Prominent and enlarged nucleoli=

basophilic mass formed of DNA & RNA, normally disappear during cell division & reappear in new daughters.

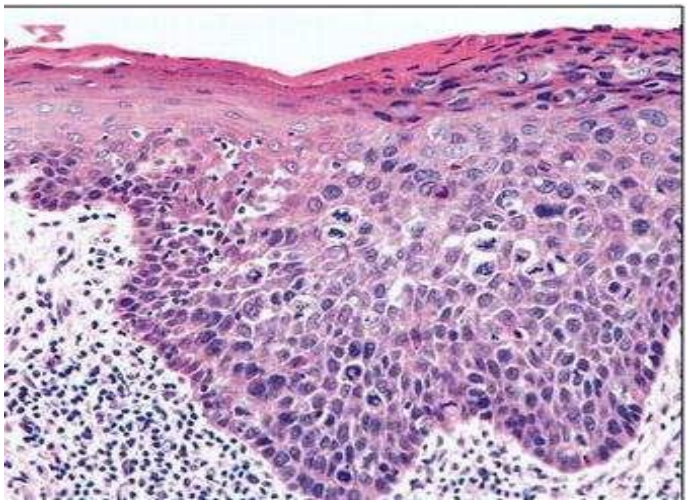


5. **Nuclear and cellular pleomorphism.** Nuclei and cells are of different size and shape. (Dyskaryosis & Piokilokarynosis)



a) **Dyskaryosis:** nuclear atypism including giant nuclei

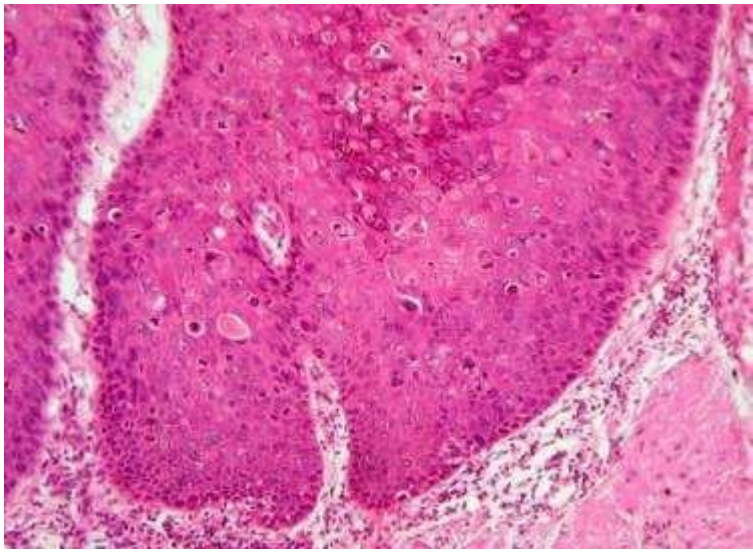
b) **Piokilokarynosis:** division of nuclei without division of cytoplasm→ **multinucleated giant cell**



c) N.B. Pleomorphism= variation in size and shape

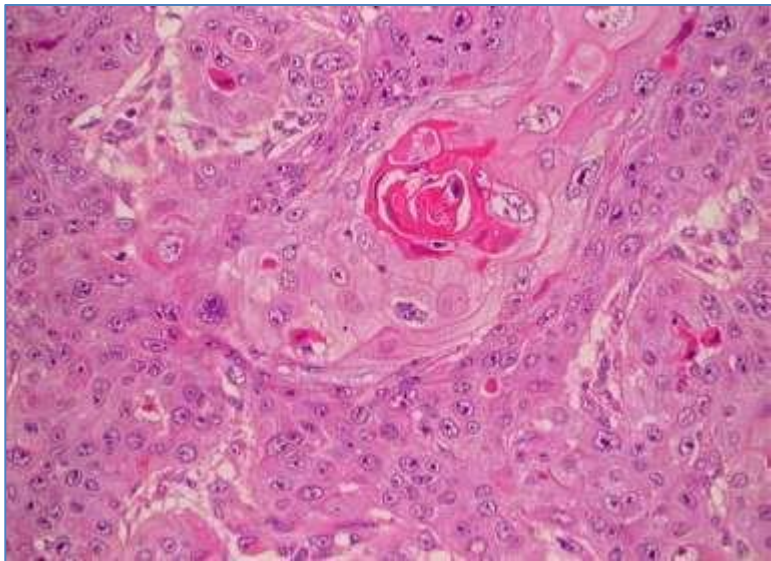
Dyskeratosis/abnormal keratinization=

- **Premature keratinization (maturation) of epithelial cells that occurring below the normal keratin layer within the spinous cell layer either:**

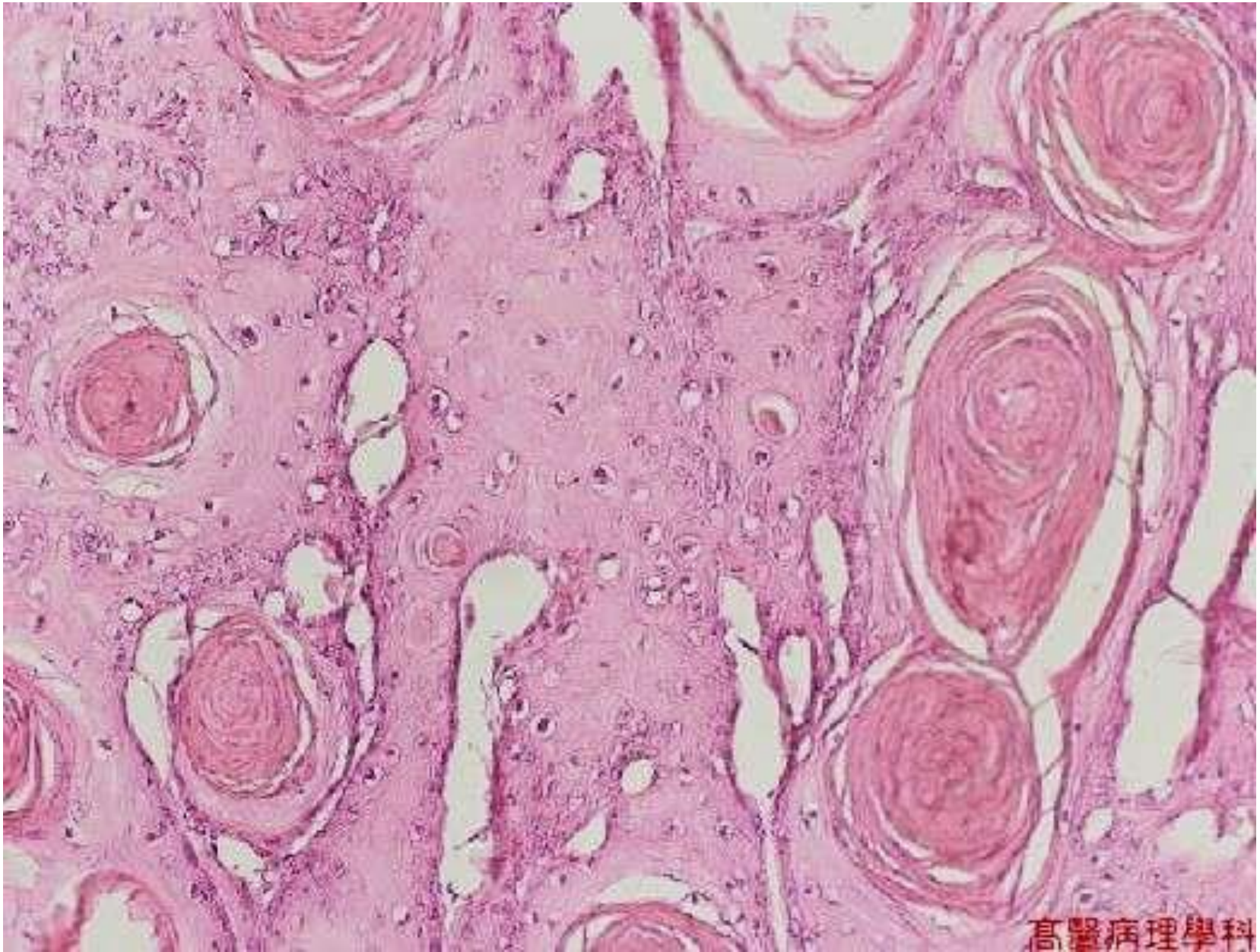


1. **Individual cell keratinization premature keratinization of individual cells within the spinous cell layer**
2. **Keratin or epithelial pearls Disturbed maturation of intraepithelial groups cells.**

(focal, round collections of concentrically layer of keratinized cells).



Keratin pearls

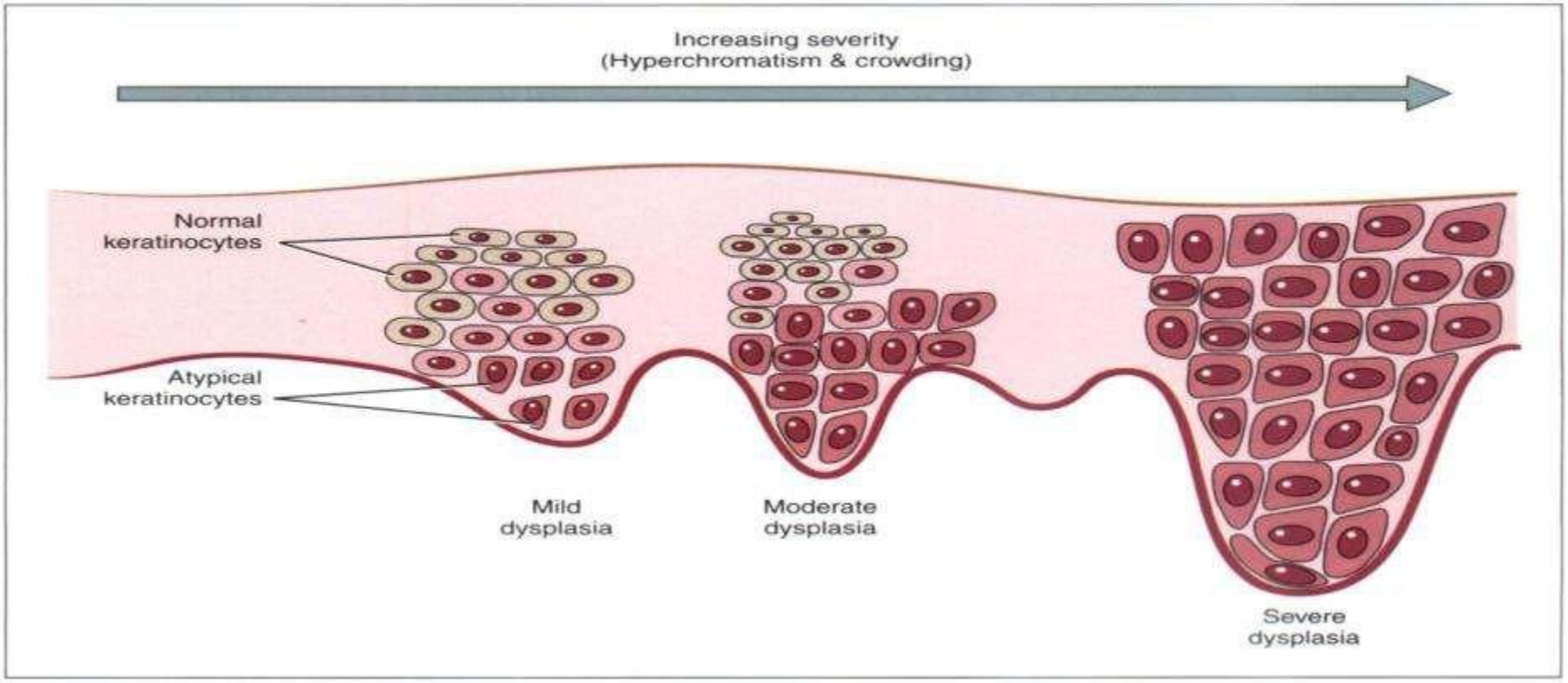


Degree/Severity of epithelial dysplasia

- ***"Severity"*** **or intensity of dysplasia are classified microscopically according to *amount of involvement of tissue* as:**

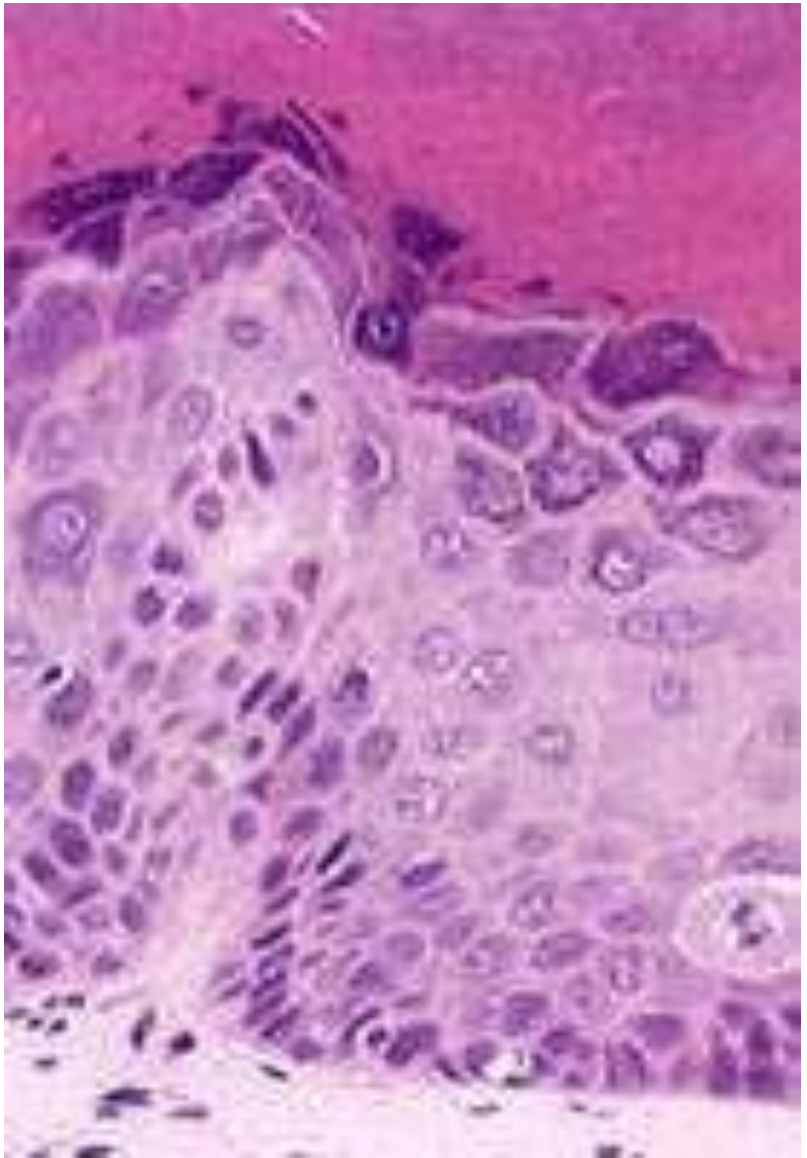
1. **Mild (focal atypia)**
2. **Moderate**
3. **Severe**

- **Mild epithelial dysplasia:** alteration limited to basal and parabasal layers.
- **Moderate epithelial dysplasia:** demonstrates involvement from basal layer to mid portion of spinous layer.
- **Severe epithelial dysplasia:** alterations from basal layer to a level above the midpoint of epithelium.

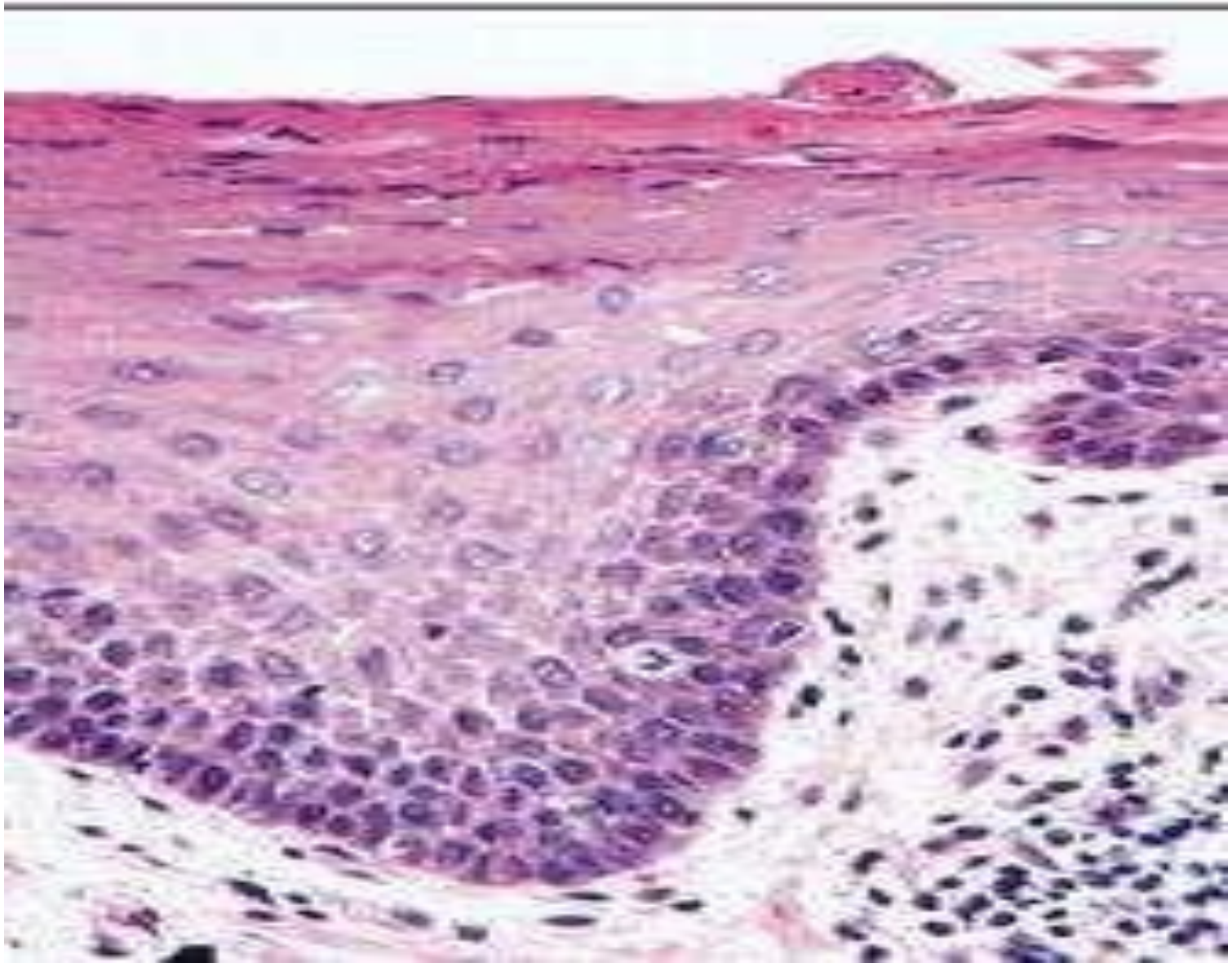


Mild epithelial dysplasia

- Alteration limited to basal and parabasal layers

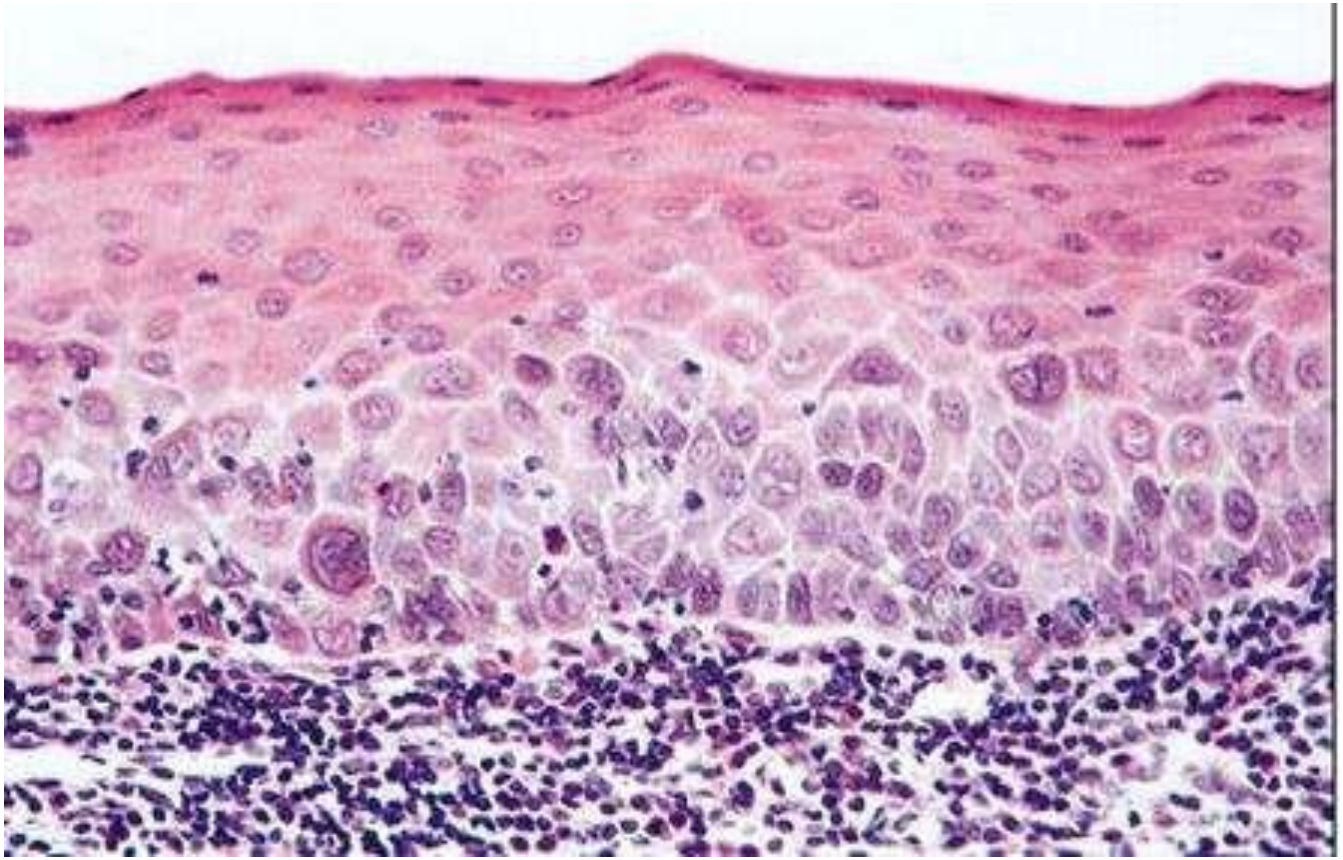


Mild epithelial dysplasia



Moderate epithelial dysplasia

- **Involvement from basal layer to mid portion of spinous layer**

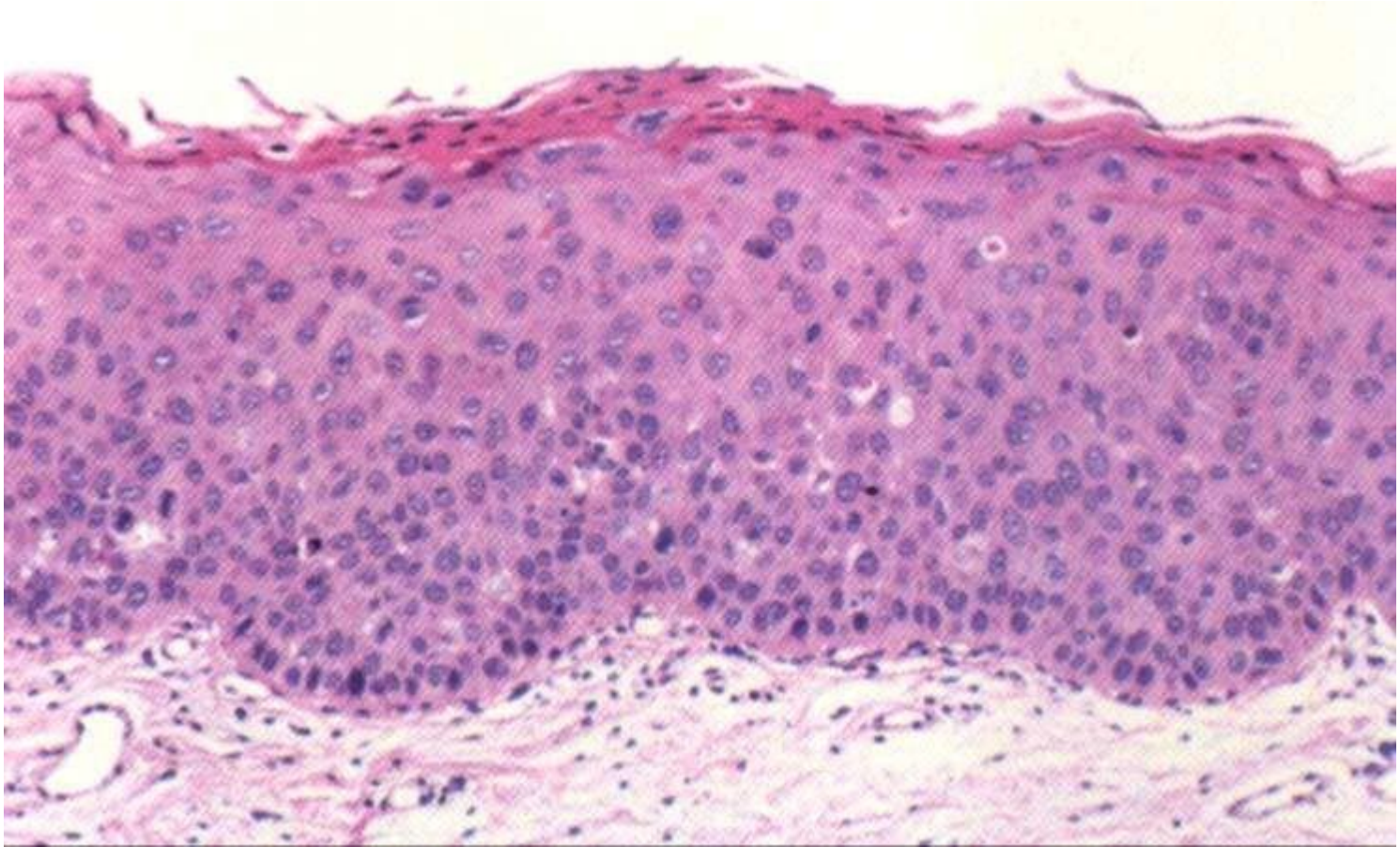


Moderate epithelial dysplasia

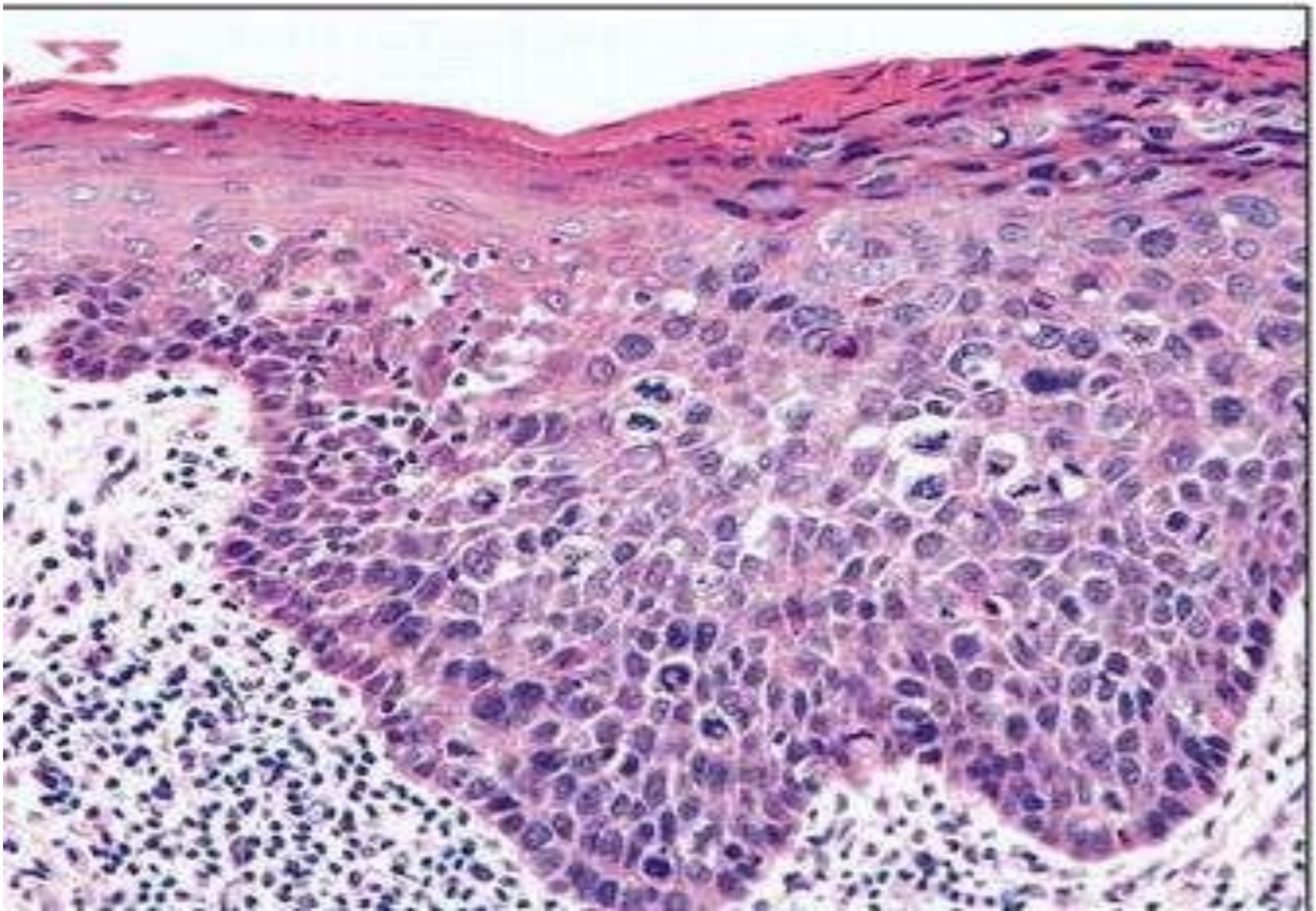


Severe epithelial dysplasia

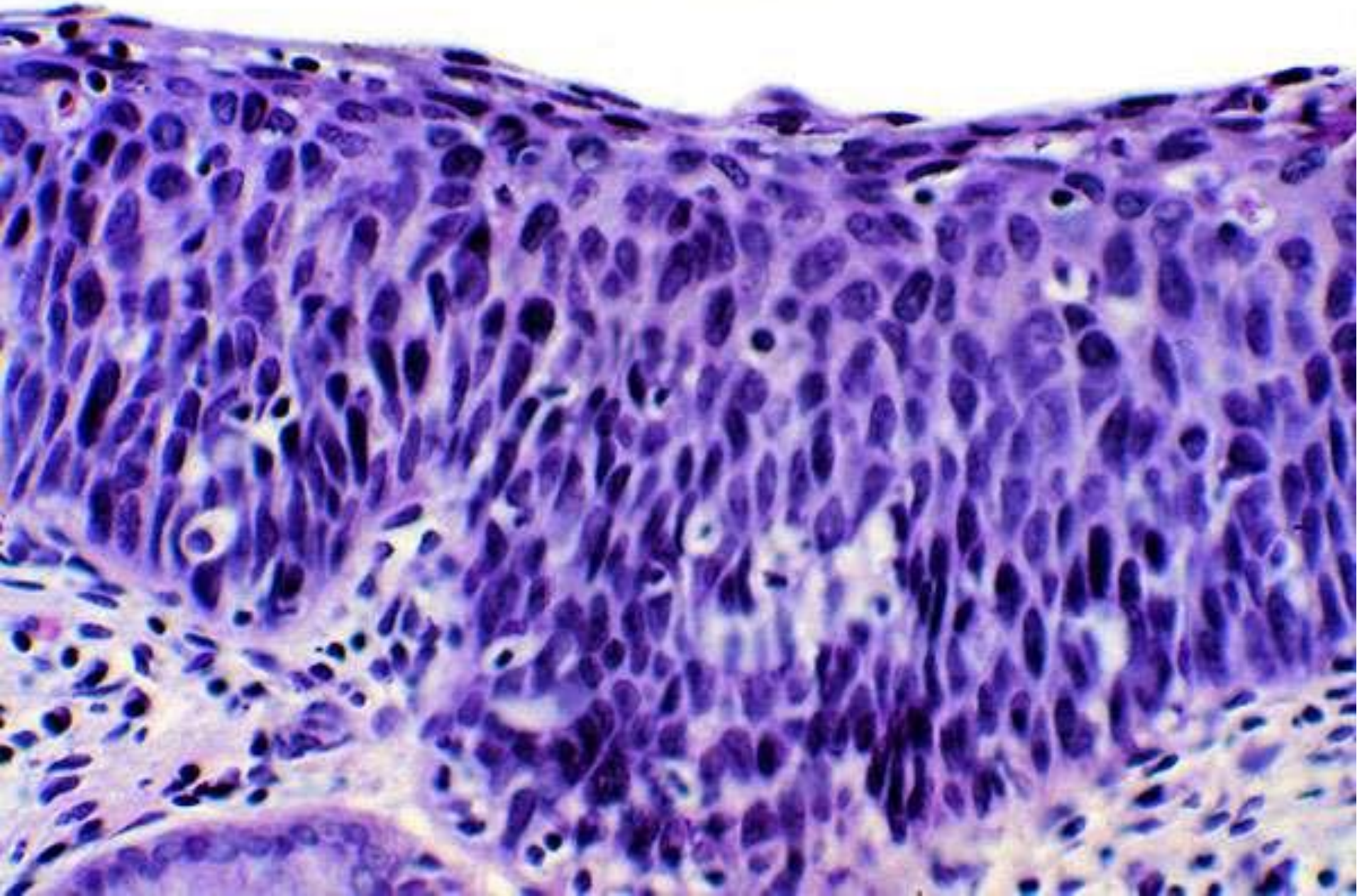
- **Alterations from basal layer to a level above the midpoint of epithelium**



Severe epithelial dysplasia

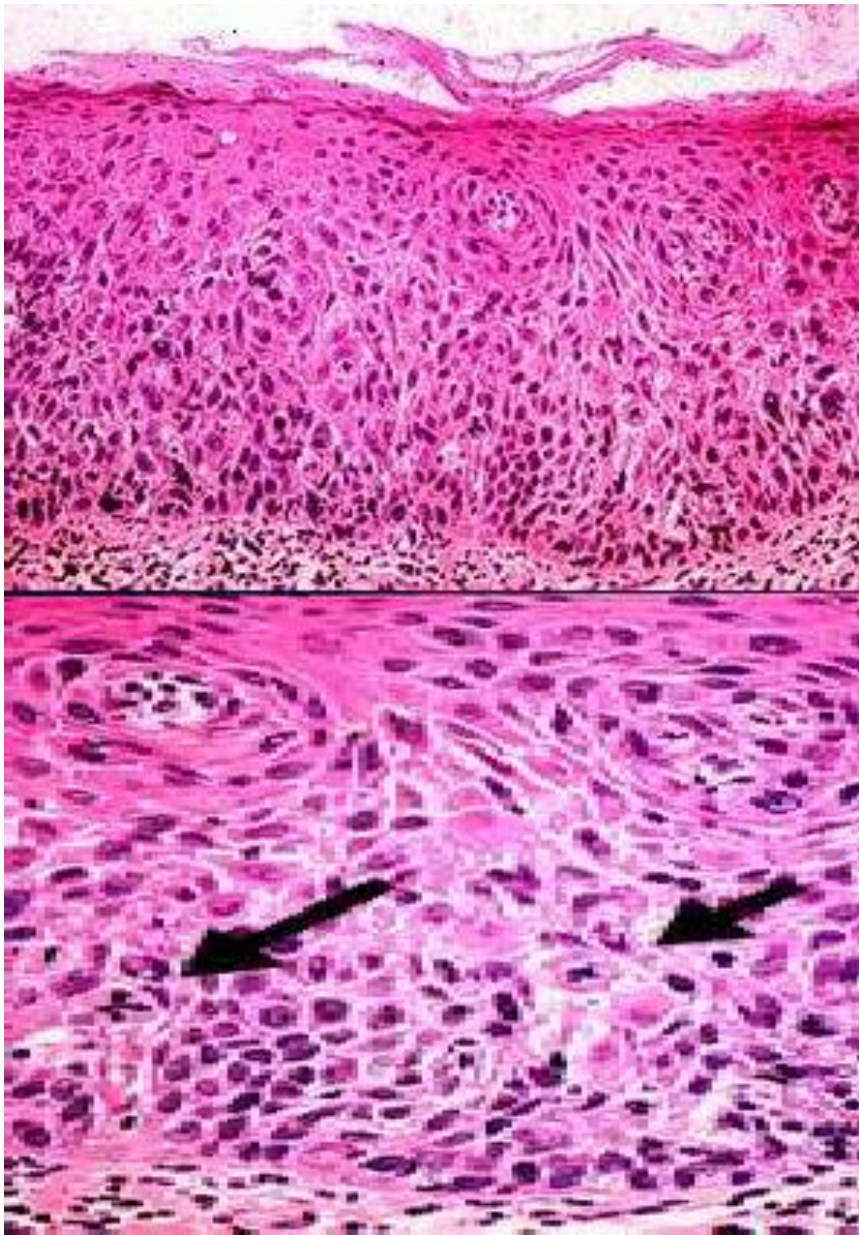


Severe epithelial dysplasia

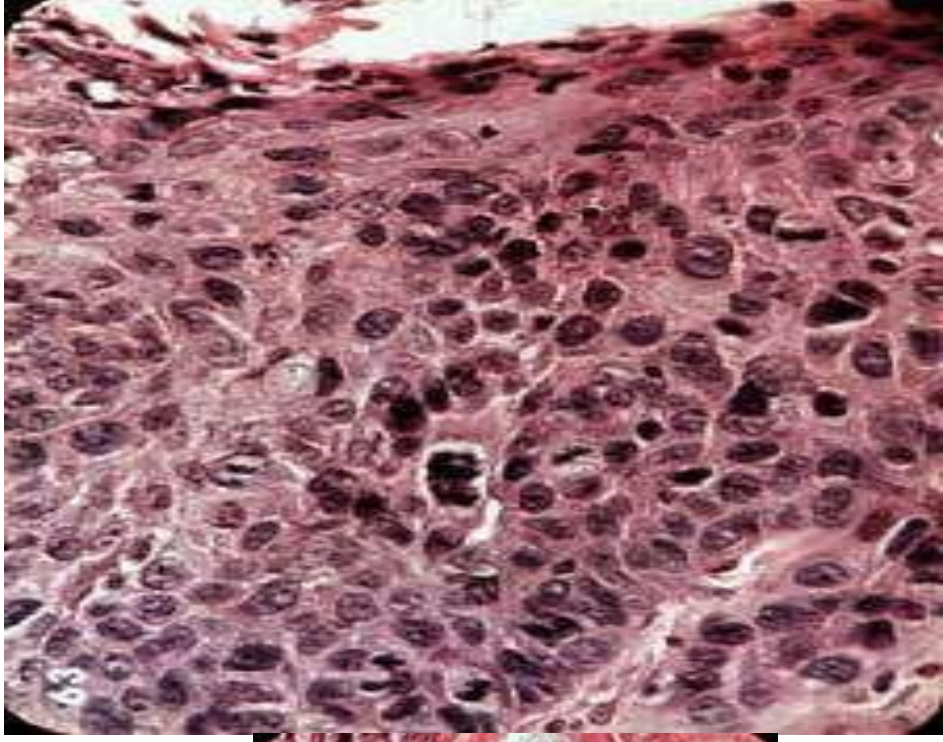


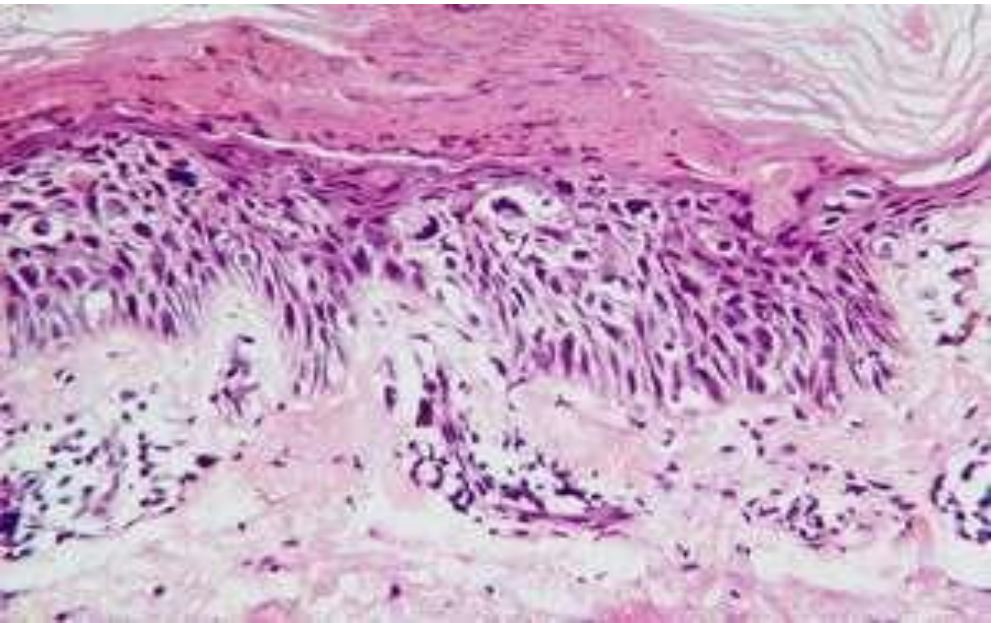
carcinoma insitu

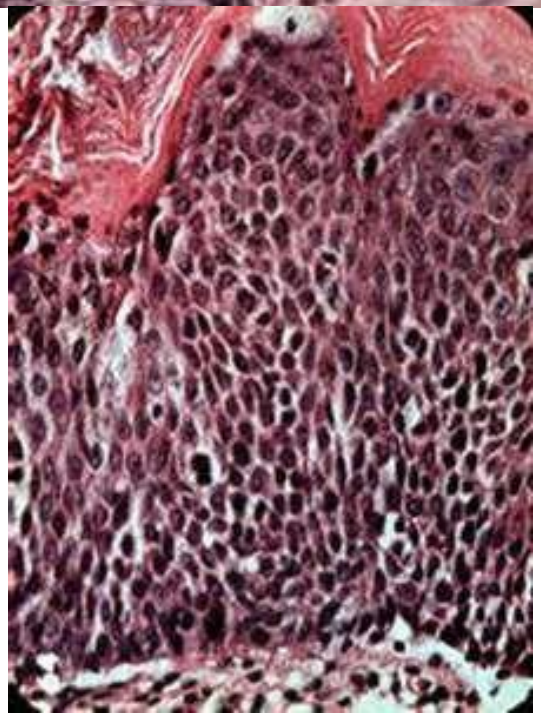
- When the entire thickness of epithelium is involved
- Carcinoma insitu → dysplastic epithelial cell that extend from the basal layer to the surface of mucosa (“top–to–bottom” change).



- The important feature of carcinoma in situ is that invasion has not occurred because the basement membrane is intact thus metastases, cannot occur.








Carcinoma *insitu*

